

Project Initialization and Planning Phase

Date	4 july,2026
Team ID	SWTID-2026-5594
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to automate the credit card approval process using Machine Learning techniques. The system analyzes an applicant's personal, financial, and employment information to predict whether the credit card application should be approved or rejected. This reduces manual effort, minimizes human errors, improves decision-making accuracy, and provides faster responses to applicants through a user-friendly Flask web application.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent Machine Learning-based Credit Card Approval Prediction System that predicts approval decisions accurately using historical applicant data. The system helps banks automate the screening process, reduce processing time, and improve operational efficiency.
Scope	.The project includes data preprocessing, feature engineering, model training, model evaluation, and deployment of the best-performing Machine Learning model through a Flask web application. The system provides real-time approval predictions and can be extended for cloud deployment using IBM Watson Machine Learning.
Problem Statement	
Description	Banks receive thousands of credit card applications every day. Manually reviewing these applications is time-consuming, expensive, and prone to human errors. Delays in approval reduce customer satisfaction and increase operational workload..
Impact	Automating the credit card approval process improves decision-making speed, reduces manual effort, minimizes errors, enhances customer experience, and helps financial institutions process applications more efficiently..
Proposed Solution	
Approach	The system uses Machine Learning algorithms such as Random Forest, Logistic Regression, Decision Tree, and XGBoost to analyze applicant information. After comparing model performance, the best-performing model is selected, saved using Joblib, and integrated into a Flask web application for real-time prediction

Key Features	• Machine Learning-based Credit Card Approval Prediction. • Real-time prediction through a Flask web application. • User-friendly interface for applicant data entry. • Fast and accurate approval or rejection prediction. • Data preprocessing and feature engineering. • Model evaluation using multiple classification algorithms. • Deployment-ready architecture with IBM Watson Machine Learning support.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Processor	CPU/GPU specifications, number of cores	intel Core i3 or above
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	Minimum 2 GB Free Disk Space
Software		
Frameworks	Python frameworks	Python 3.8+
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv